

Class 10: Halloween Mini-Project

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Background

As it is nearly Halloween and the half way point in the quarter, let's do a mini project that help us figure out the best candy!

Our data come from the 538 website and is available as a CSV file.

Data Import

```
candy <- read.csv(file="candy-data.txt",row.names=1)
flextable::flextable(head(candy,10))
```

chocolate	fruity	caramel	peanutyalmond	nougat	crispedrice	wafer	hard	bar	pluribus s
1	0	1	0	0	1	0	0	1	0
1	0	0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0

chocolate	fruity	caramel	peanut	almond	nougat	crisp	rice wafer	hard	bar	pluribus s
0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	1	0
1	0	1	1	1	1	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0	0	0	1
0	1	1	0	0	0	0	0	0	0	0

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

There are 85 candy types in this dataset.

Q2. How many fruity candy types are in the dataset?

```
library(tidyverse)
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in the dataset.

Q3. What is your favorite candy in the dataset and what is its winpercent value?

```
candy %>%
  filter(rownames(candy) == "Kit Kat") %>%
  select(winpercent)
```

```
      winpercent
Kit Kat    76.7686
```

My favorite candy is Kit Kat and its win rate is 76.77%.

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat",]$winpercent
```

```
[1] 76.7686
```

The winpercent value for Kit Kat is 76.77%

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy%>%  
  filter(rownames(candy)=="Tootsie Roll Snack Bars")%>%  
  select(winpercent)
```

	winpercent
Tootsie Roll Snack Bars	49.6535

The winpercent for Toosie Roll Snack Bars is 49.65%.

Quick Overview of Data

```
skimr::skim(candy)
```

Table 2: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

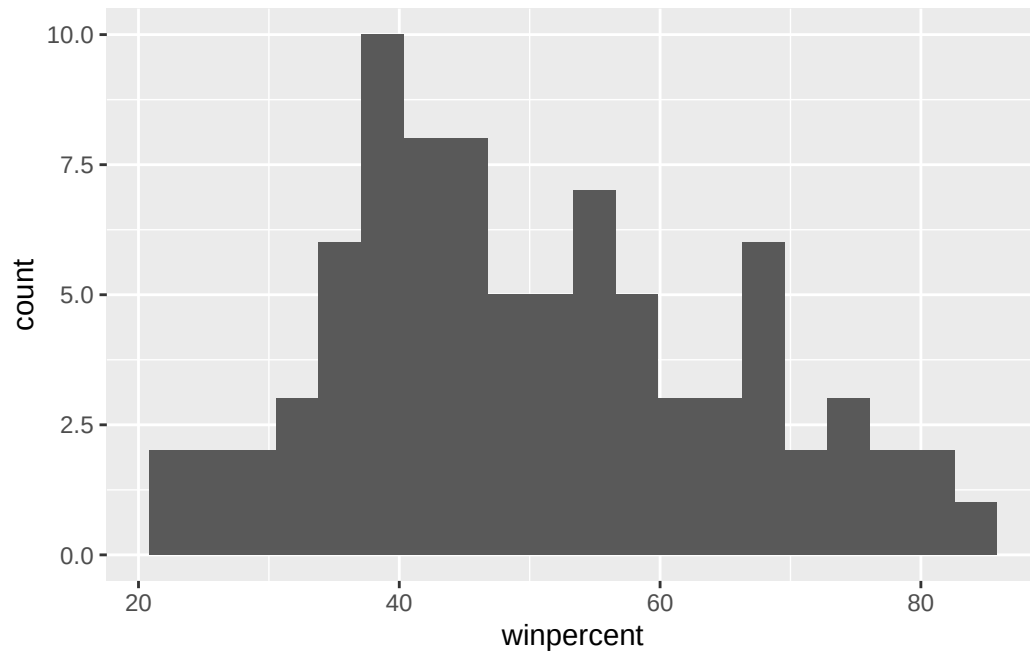
The winpercent is on a 0-100 scale and the rest are on 0-1 scale

Q7. What do you think a zero and one represent for the candy\$chocolate column?

The candy does not contain chocolate.

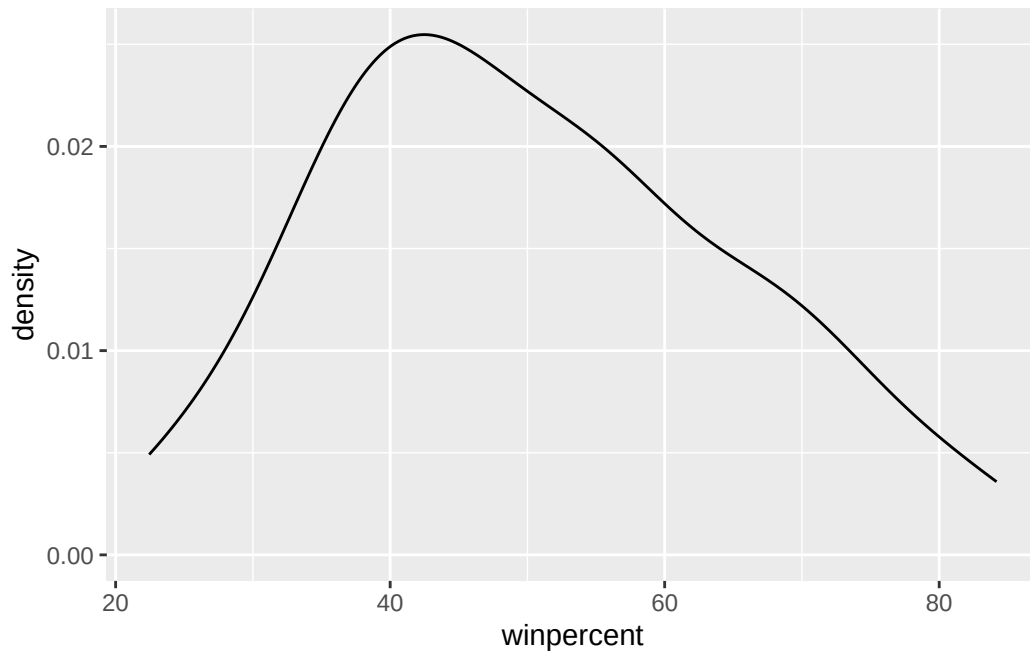
Q8. Plot a histogram of winpercent values

```
ggplot(candy)+
  aes(winpercent)+
  geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

```
ggplot(candy)+  
  aes(winpercent)+  
  geom_density()
```



The distribution is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The center of the distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# 1. Find all chocolate candy
# 2. Find their winpercent
# 3. Calculate the mean
# 4-6 Repeat with fruit candy

choc.win <- candy%>%
  filter(chocolate==1)%>%
  select(winpercent)

choc.mean <- mean(choc.win$winpercent)
```

```
friut.win <- candy%>%
  filter(fruity==1)%>%
  select(winpercent)

fruit.mean <- mean(friut.win$winpercent)
```

Q12. Is this difference statistically significant?

```
t.test(choc.win,friut.win)
```

Welch Two Sample t-test

```
data:  choc.win and friut.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent)|>
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325

Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511
	winpercent					
Nik L Nip	22.44534					
Boston Baked Beans	23.41782					
Chiclets	24.52499					
Super Bubble	27.30386					
Jawbusters	28.12744					

The five least liked candy types are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

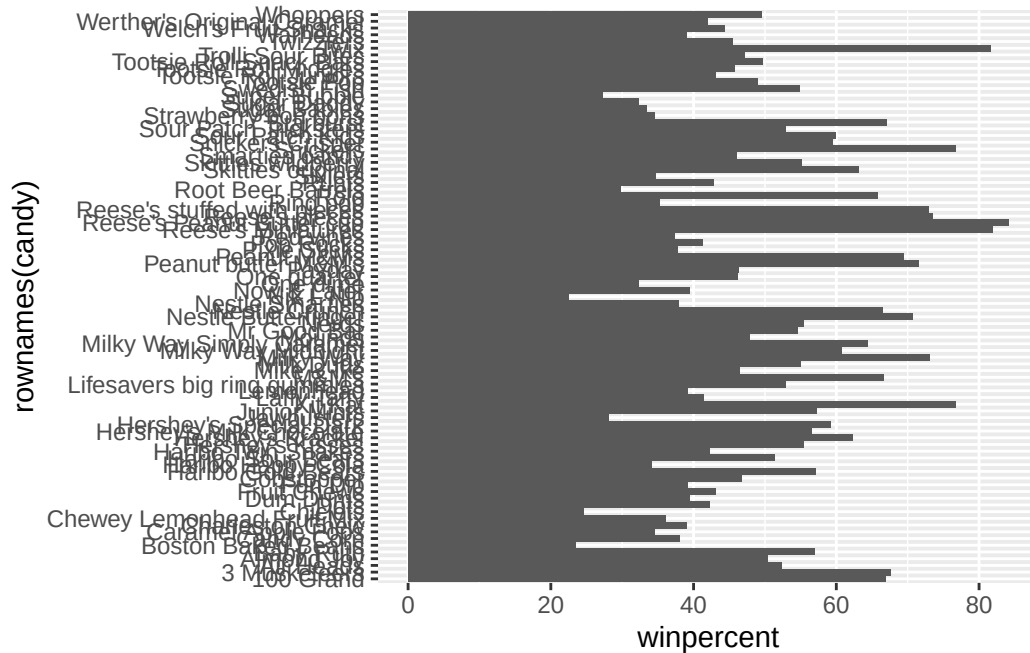
```
candy|>
  arrange(-winpercent)|>
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1
	crisped	rice	wafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0	0.720
Reese's Miniatures		0	0	0	0	0.034
Twix		1	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Snickers		0	0	1	0	0.546
	pricepercent	winpercent				
Reese's Peanut Butter cup	0.651	84.18029				
Reese's Miniatures	0.279	81.86626				
Twix	0.906	81.64291				
Kit Kat	0.511	76.76860				
Snickers	0.651	76.67378				

The top five most popular candies are Reese's Peanut Butter Cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

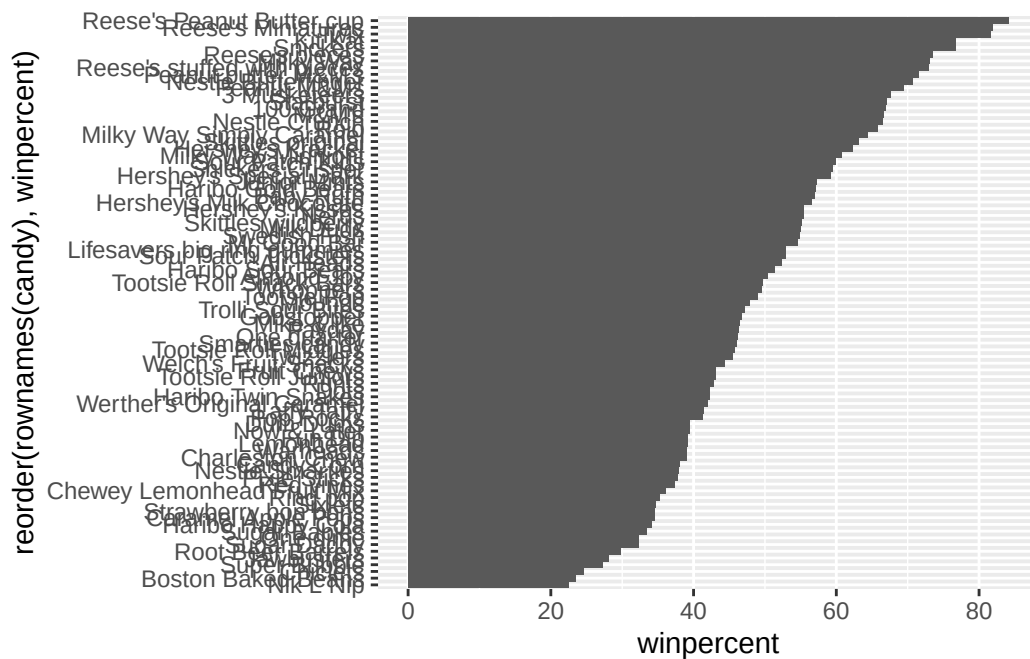
Q15. Make a first barplot of candy ranking based on winpercent values.


```
ggplot(candy)+
  aes(winpercent,rownames(candy))+
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

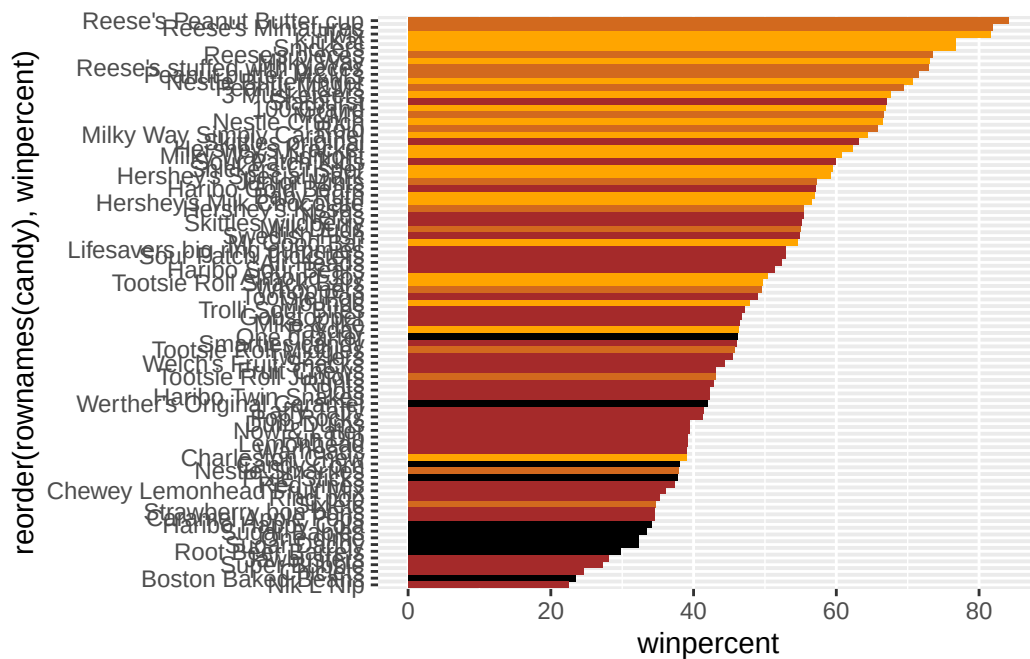
```
ggplot(candy)+
  aes(x=winpercent,
      y=reorder(rownames(candy),winpercent))+
  geom_col()
```



Add some color based on the “type of candy”

```
my_cols <- rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] <- "chocolate"
my_cols[as.logical(candy$bar)] <- "orange"
my_cols[as.logical(candy$fruity)] <- "brown"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols)
```

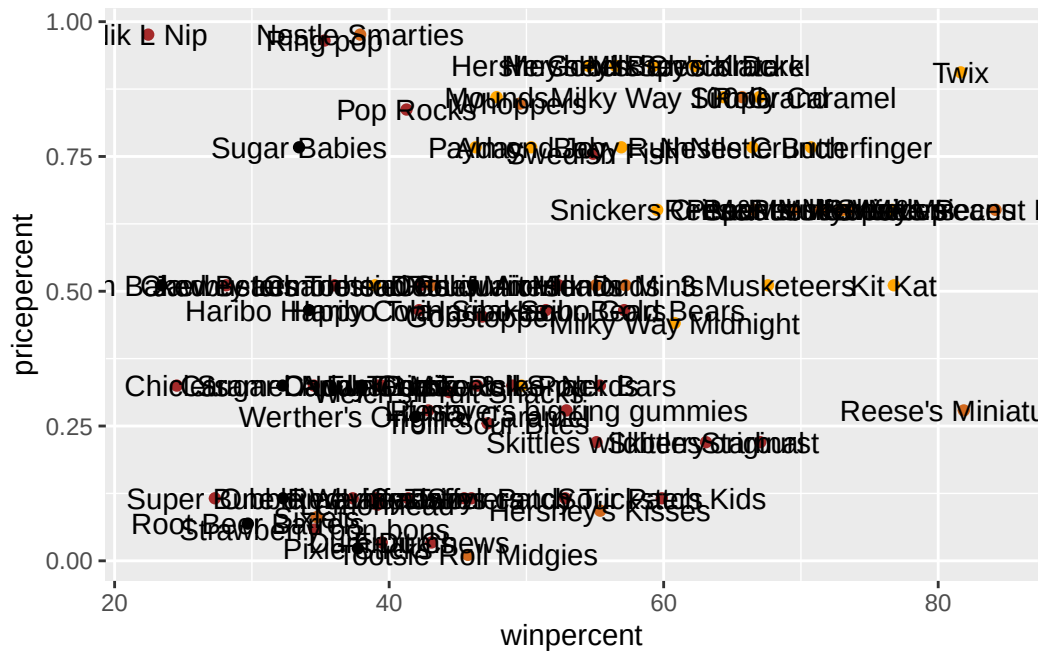


Winpercent and Pricepercent

A plot with both variables/columns winpercent and pricepercent

```
my_cols[as.logical(candy$fruity)] <- "brown"

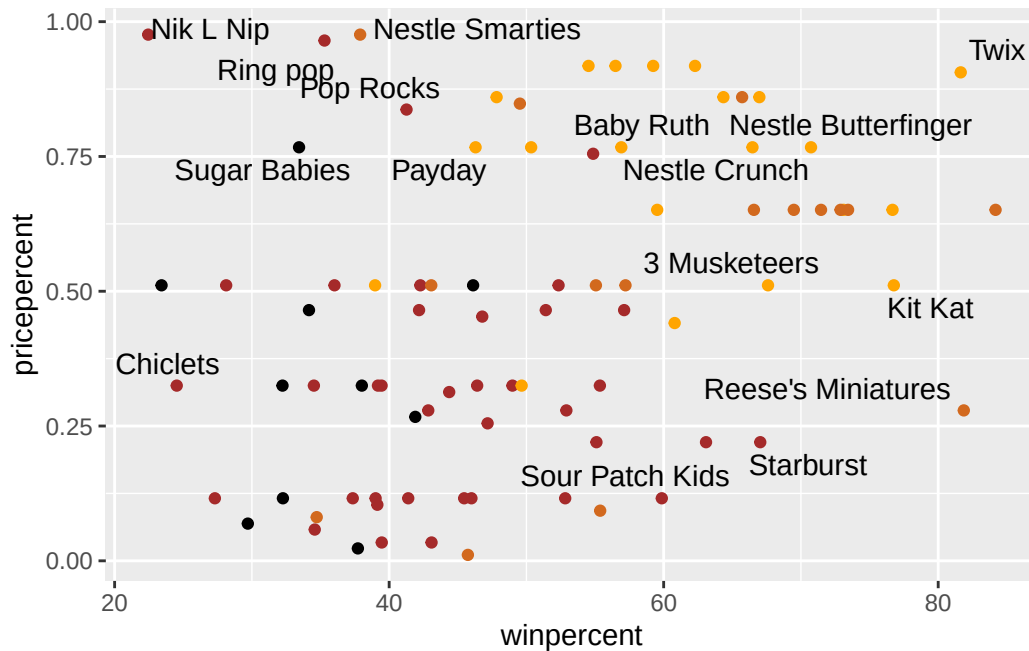
ggplot(candy)+
  aes(winpercent,pricepercent,
      label=rownames(candy))+
  geom_point(col=my_cols)+
  geom_text()
```



```
library(ggrepel)

ggplot(candy)+
  aes(winpercent,pricepercent,
      label=rownames(candy))+
  geom_point(col=my_cols)+
  geom_text_repel(max.overlaps = 6)
```

Warning: ggrepel: 69 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is sixlets.

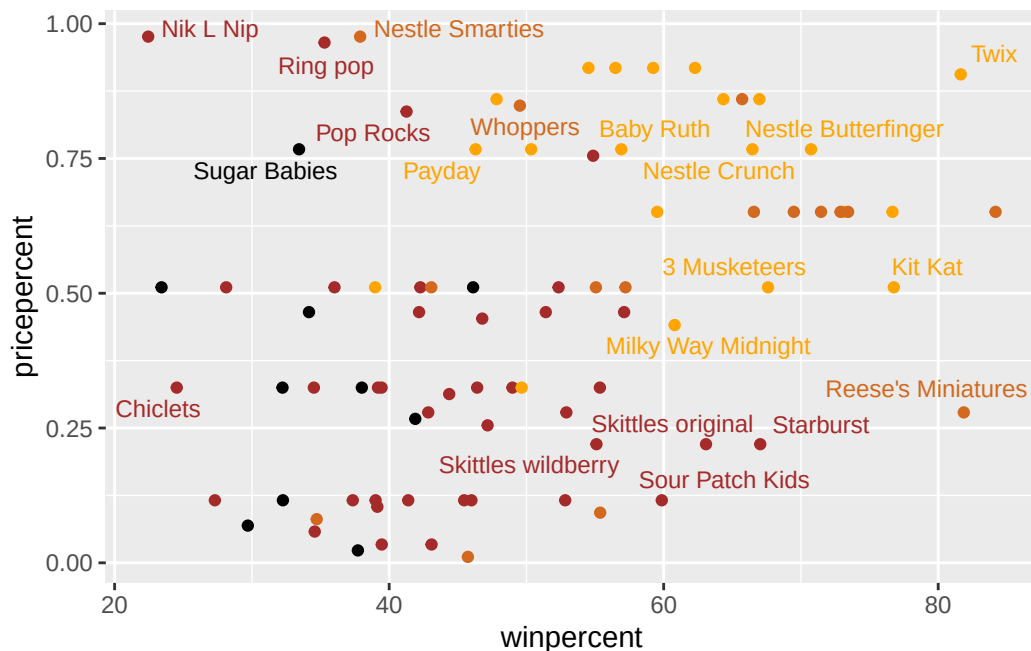
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is starburst.

Taking a look at pricepercent

```
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures is ranked highest for winpercent and lowest for pricepercent.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

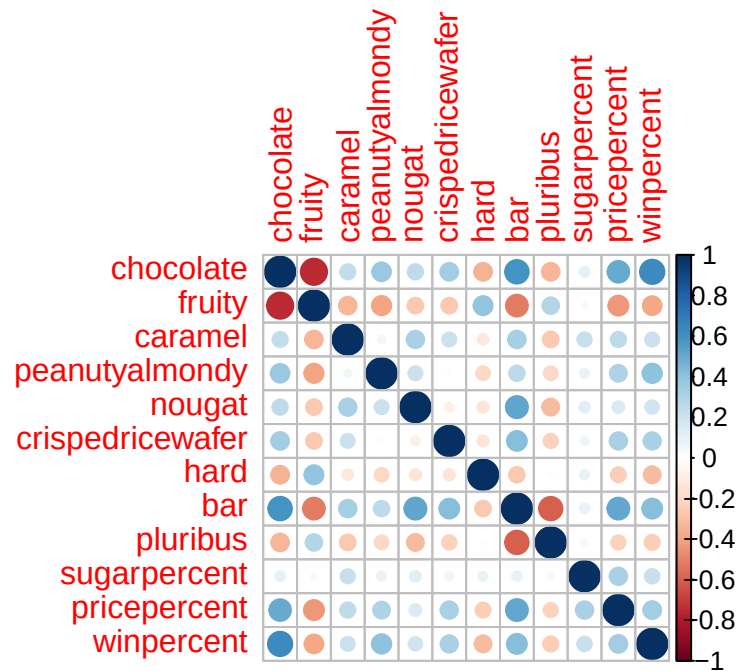
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive candy that are least popular are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate.

Exploring the correlation structure

Now that we've explored the dataset a little, we'll see how the variables interact with one another. We'll use correlation and view the results with the `corrplot` package to plot a correlation matrix.

```
library(corrplot)
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and Chocolate are anti-correlated.

Q23. Similarly, what two variables are most positively correlated?

Chocolate and Bar are most positively correlated.

Principal Component Analysis

The function to use is called `prcomp()` with an optional `scale=T/F` argument.

```
pca <- prcomp(candy,scale=T)
summary(pca)
```

Importance of components:

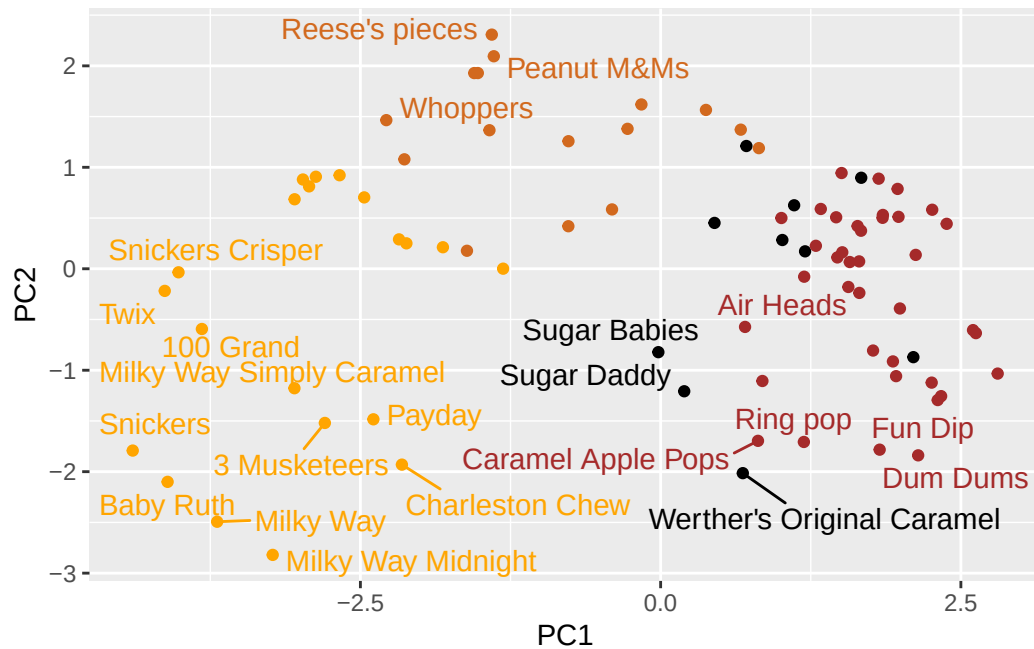
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

Our main PCA result figure

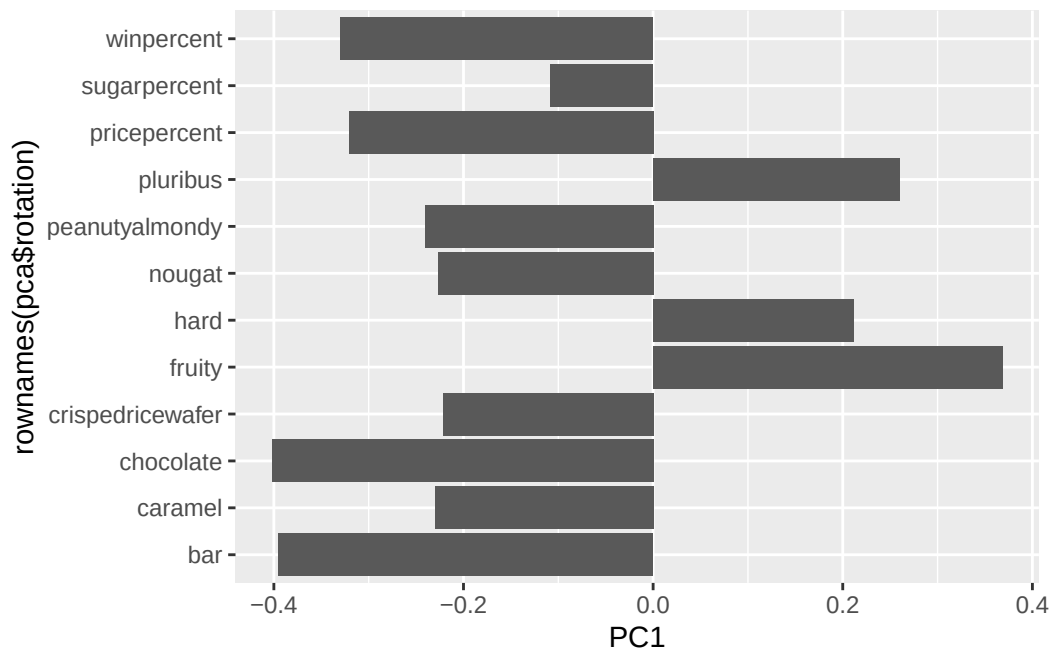
```
ggplot(pca$x)+
  aes(PC1,PC2,
       label=rownames(pca$x))+
  geom_point(col=my_cols)+
  geom_text_repel(col=my_cols, max.overlaps = 7)
```

Warning: ggrepel: 63 unlabeled data points (too many overlaps). Consider increasing max.overlaps



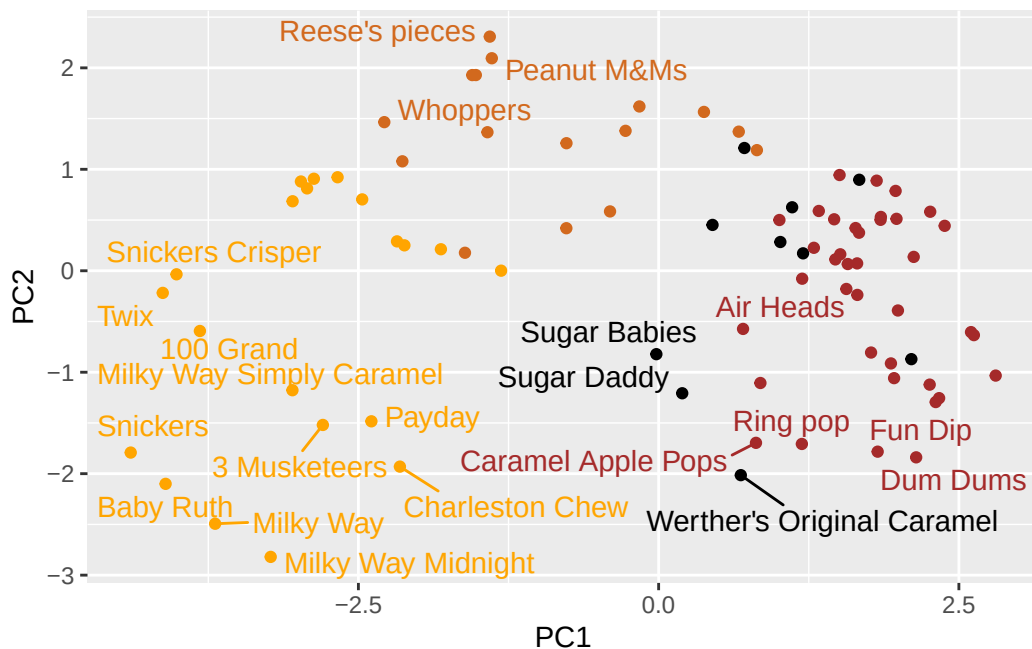
We should also examine the “variable loading” or contributions of the original variables to the new PCs.

```
ggplot(pca$rotation)+
  aes(PC1,rownames(pca$rotation))+
  geom_col()
```



```
p <- ggplot(pca$x)+
  aes(PC1,PC2,
       label=rownames(pca$x))+
  geom_point(col=my_cols)+
  geom_text_repel(col=my_cols, max.overlaps = 7)
p
```

Warning: ggrepel: 63 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Interactive plots that can be zoomed in and “brushed” over can be made with the **plotly** package. Its output is interactive and will not render to PDF.

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

```
last_plot
```

The following object is masked from 'package:stats':

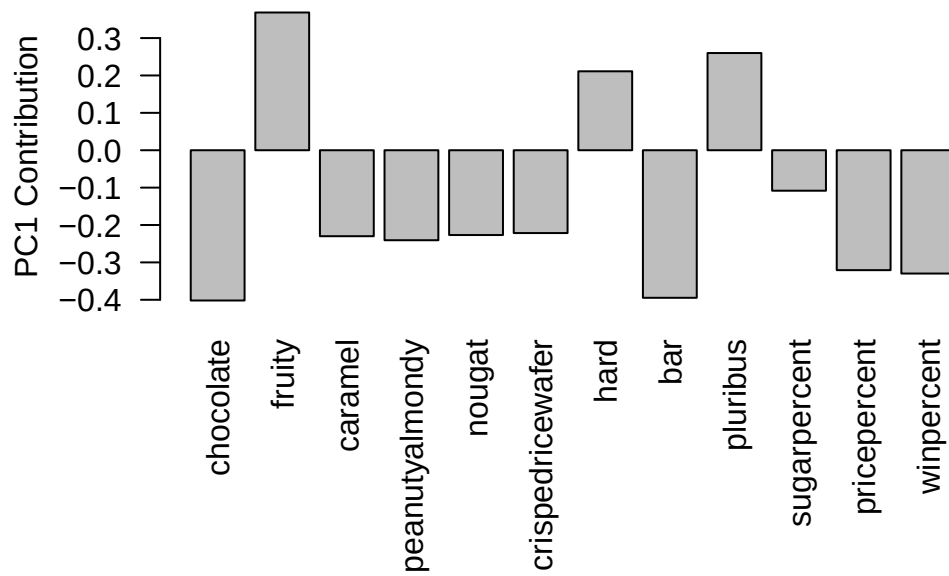
```
filter
```

The following object is masked from 'package:graphics':

```
layout
```

```
#ggplotly(p)
```

```
par(mar=c(8,4,2,2))  
barplot(pca$rotation[,1], las=2, ylab="PC1 Contribution")
```



Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

Fruity, hard, pluribus are picked up strongly by PC1 in the positive direction. This makes sense because candies with these three characteristics tend to correlate less with candies of other characteristics.